

OpenReview export (answer3.tex)

We thank the reviewer for the careful reading and thoughtful questions.

Q1 (Technical novelty). We apologize for not being precise enough in the coupling argument presentation: one intermediate step was missing in the draft wording.

More precisely, there is no technical novelty for the existence/uniqueness proof following Nguyen et al. The only technical novelty comes from modifying the coupling template to our channel-wise sum (their three-layer assumption / coupling display): after introducing the coupling π_t and using bounded first-layer activation, one obtains a bound of the form

$$\mathbb{E}_{\pi_t} \left[\left| \langle \mathbb{E}_Z [\Delta_2^{H*}(t, \cdot) - \Delta_2^{H*}(\cdot; \bar{w}) \mid X = x], \varphi_1(\langle u, x \rangle) \rangle \right| \right] \leq K \mathbb{E}_{\pi_t} \left[|\Delta_2^{H*}(t, \cdot) - \Delta_2^{H*}(\cdot; \bar{w})| \right],$$

and one then controls the right-hand side by weighted coupled gaps (same logic as Nguyen & Pham), with low-rank-specific channel aggregation.

Our low-rank translation of this missing step is that the main coupling term is controlled through a channel sum under π_t :

$$\mathbb{E}_{\pi_t} \left[(1 + |\bar{w}_2|) |\bar{w}_2| \sum_{k=1}^r |\bar{w}_{1,k}| |w_{1,k} - \bar{w}_{1,k}| \right] \rightarrow 0,$$

with notation matched to our neuronal embedding variables in Eq. (1). This is exactly the place where the RF-LR structure replaces the fully connected contraction chain by channelwise control plus mixing bounds.

We will make this explicit in the appendix by adding: (i) the *post-translates-to* display; (ii) the time-regularity condition for each channel,

$$\operatorname{ess\,sup}_t |\partial_t w_1(t, U_k)| \rightarrow 0, \quad k = 1, \dots, r;$$

(iii) the bi-Lipschitz/bounded-mixing constants used to close the backward estimate.

For ReLU, the closure step uses the sharp inequality

$$\left| \operatorname{ReLU} \left(\sum_{i=1}^r x_i \right) - \operatorname{ReLU} \left(\sum_{i=1}^r y_i \right) \right| \leq \left| \sum_{i=1}^r (x_i - y_i) \right| \leq \sum_{i=1}^r |x_i - y_i|$$

rather than any extra factor-of- r slack. We will state this explicitly (with compact-support / boundedness assumptions for pre-activations) to clarify why r -channel coupling closes.

Q2 (Spectral bias and Section 4). On universality and training regimes: we will revise Section 4 to separate statements cleanly. In the NTK / lazy training regime, the effective bias is governed by the kernel spectrum and one expects the classical low-frequency-first behavior emphasized in that line of work. The empirical “tilt” toward comparatively higher-frequency structure that we highlight is observed in a richer feature-learning setting for our low-rank RF architecture; we do not claim that this behavior is universal across all scalings (including finite width outside mean-field parameterizations or lazy limits).

Regarding Rahaman et al., “On the Spectral Bias of Neural Networks” (2019), arXiv:1806.08734, <https://arxiv.org/abs/1806.08734>, is an apt reference for the standard Fourier / low-frequency-first picture in sufficiently wide, kernel-like regimes.

On why low-rank maps might carry more high-frequency content (strictly heuristic, work in progress / under review): in depth- L ReLU networks, one can view NN outputs through “sum over paths” (per-region linear parametrizations); shrinking the effective rank reduces how many independent hinge directions can co-activate across depth, which loosely relaxes constraints on how linear regions can tile input space—informally, a less constrained CPWL can allow relatively

more high-dimensional faces versus low-dimensional ones and thus alter how energy is distributed across spatial frequencies in a Fourier lens. This complements the shorter “fewer effectively independent hinge directions / boundary geometry” phrasing we will keep in the main text and will be marked explicitly as exploratory reasoning and ongoing review, not as a formal implication.

On exposition and scope: we agree Section 4 is currently qualitative and that the two-point analytic anchor is minimal. We will tone down claims, acknowledge the simplicity of the two-point construction, and add citations for the CPWL / compositional picture and Fourier heuristics beyond Rahaman et al.